

Layer 03 VLAN,SVI & HSRP Configuration LAB – CCNA

Floor-SW-01

```
Floor-SW-01>enable
Floor-SW-01#conf terminal
Floor-SW-01(config)#hostname Floor-SW-01
Floor-SW-01(config)#vlan 2
Floor-SW-01(config-vlan)#name LAB-01
Floor-SW-01(config-vlan)#exit
Floor-SW-01(config)#interface range fastEthernet 0/1-5
Floor-SW-01(config-if-range)#switchport mode access
Floor-SW-01(config-if-range)#switchport access vlan 2
Floor-SW-01(config-if-range)#exit
Floor-SW-01(config)#vlan 3
Floor-SW-01(config-vlan)#name LAB-02
Floor-SW-01(config-vlan)#exit
Floor-SW-01(config)#interface range fastEthernet 0/6-10
Floor-SW-01(config-if-range)#switchport mode access
Floor-SW-01(config-if-range)#switchport access vlan 3
Floor-SW-01(config-if-range)#exit
Floor-SW-01(config)#vlan 4
Floor-SW-01(config-vlan)#name LAB-03
Floor-SW-01(config-vlan)#exit
Floor-SW-01(config)#interface range fastEthernet 0/11-15
Floor-SW-01(config-if-range)#switchport mode access
Floor-SW-01(config-if-range)#switchport access vlan 4
Floor-SW-01(config-if-range)#exit
Floor-SW-01(config)#exit
Floor-SW-01(config)#interface range gigabitEthernet 0/1-2
Floor-SW-01(config-if-range)#switchport mode trunk
Floor-SW-01(config-if-range)#exit
```

Floor-SW-02

```
Switch>enable
Switch#conf t
Switch(config)#vlan 2
Switch(config-vlan)#name LAB-01
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/1-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name LAB-02
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/6-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
Switch(config-if-range)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name LAB-03
Switch(config-vlan)#exit
Switch(config)#interface range fastEthernet 0/11-15
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 4
Switch(config-if-range)#exit
Switch(config)#interface range gigabitEthernet 0/1-2
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#exit

Switch(config)#hostname Floor-SW-02
Floor-SW-02(config)#exit
```

Floor-SW-03

Switch>enable

Switch#conf t

Switch(config)#hostname Floor-SW-03

Floor-SW-03(config)#vlan 2

Floor-SW-03(config-vlan)#name LAB-01

Floor-SW-03(config-vlan)#exit

Floor-SW-03(config)#interface range fastEthernet 0/1-5

Floor-SW-03(config-if-range)#switchport mode access

Floor-SW-03(config-if-range)#switchport access vlan 2

Floor-SW-03(config-if-range)#exit

Floor-SW-03(config)#vlan 3

Floor-SW-03(config-vlan)#name LAB-02

Floor-SW-03(config-vlan)#exit

Floor-SW-03(config)#interface range fastEthernet 0/6-10

Floor-SW-03(config-if-range)#switchport mode access

Floor-SW-03(config-if-range)#switchport access vlan 3

Floor-SW-03(config-if-range)#exit

Floor-SW-03(config)#vlan 4

Floor-SW-03(config-vlan)#name LAB-03

Floor-SW-03(config-vlan)#exit

Floor-SW-03(config)#interface range fastEthernet 0/11-15

Floor-SW-03(config-if-range)#switchport mode access

Floor-SW-03(config-if-range)#switchport access vlan 4

Floor-SW-03(config-if-range)#exit

Floor-SW-03(config)#interface range gigabitEthernet 0/1-2

Floor-SW-03(config-if-range)#switchport mode trunk

Floor-SW-03(config-if-range)#exit

Floor-SW-03(config)#exit

Floor-SW-03#wr

Building configuration...

[OK]

ML-SW-01

Switchenable

SwitchAconfigure terminal

Switch(config)#hostname MLSW-01

interface gigabitEthernet 1/0/1

description FLOOR-SW-01-Up-Link

switchport mode trunk

exit

interface gigabitEthernet 1/0/2

description FLOOR-SW-02-Up-Link

switchport mode trunk

exit

interface gigabitEthernet 1/0/3

description FLOOR-SW-03-Up-Link

switchport mode trunk

exit

ML-SW-01(config)#vlan 2

ML-SW-01(config-vlan)#name LAB-01

ML-SW-01(config-vlan)#exit

ML-SW-01(config)#vlan 3

ML-SW-01(config-vlan)#name LAB-02

ML-SW-01(config-vlan)#exit

ML-SW-01(config)#vlan 4

ML-SW-01(config-vlan)#name LAB-03

ML-SW-01(config-vlan)#exit

ML-SW-01(config)#vlan 3

ML-SW-01(config-vlan)#name LAB-02

ML-SW-01(config-vlan)#exit

ML-SW-01(config)#vlan 4

ML-SW-01(config-vlan)#name LAB-03

ML-SW-01(config-vlan)#exit

```
ML-SW-01(config)#interface gigabitEthernet 1/0/24
ML-SW-01(config-if)#description ISP-UP-Link
ML-SW-01(config-if)#no switchport
ML-SW-01(config-if)#ip address 100.100.10.2 255.255.255.252
ML-SW-01(config-if)#no shutdown
ML-SW-01(config-if)#exit
ML-SW-01(config)#interface vlan 2
ML-SW-01(config-if)#description VLAN-02-LAB-01
ML-SW-01(config-if)#ip address 192.168.2.2 255.255.255.0
ML-SW-01(config-if)#description VLAN-02-LAB-01
ML-SW-01(config-if)#standby 2 preempt
ML-SW-01(config-if)#standby 2 priority 150
ML-SW-01(config-if)#standby 2 ip 192.168.2.1
ML-SW-01(config-if)#no shutdown
ML-SW-01(config-if)#exit
ML-SW-01(config)#interface vlan 3
ML-SW-01(config-if)#description VLAN-03-LAB-02
ML-SW-01(config-if)#ip address 192.168.3.2 255.255.255.0
ML-SW-01(config-if)#standby 2 preempt
ML-SW-01(config-if)#standby 2 priority 150
ML-SW-01(config-if)#standby 2 ip 192.168.3.1
ML-SW-01(config-if)#no shutdown
ML-SW-01(config-if)#exit
ML-SW-01(config)#interface vlan 4
ML-SW-01(config-if)#description VLAN-04-LAB-03
ML-SW-01(config-if)#ip address 192.168.4.2 255.255.255.0
ML-SW-01(config-if)#standby 2 preempt
ML-SW-01(config-if)#standby 2 priority 150
ML-SW-01(config-if)#standby 2 ip 192.168.4.1
ML-SW-01(config-if)#no shutdown
ML-SW-01(config-if)#exit
ML-SW-01(config)#
```

ML-SW-01(config)#

ML-SW-01(config)#ip routing

ML-SW-01(config)#ip route 0.0.0.0 0.0.0.0 100.100.10.1

ML-SW-01(config)#exit

ML-SW-01#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

ML-SW-01#

ML-SW-02

```
hostname MLSW-02
interface gigabitEthernet 1/0/1
description FLOOR-SW-01-Up-Link
switchport mode trunk
exit
interface gigabitEthernet 1/0/2
description FLOOR-SW-02-Up-Link
switchport mode trunk
exit
interface gigabitEthernet 1/0/3
description FLOOR-SW-03-Up-Link
switchport mode trunk
exit
vlan 2
name LAB-01
exit
vlan 3
name LAB-02
exit
vlan 4
name LAB-03
exit
interface gigabitEthernet 1/0/24
description ISP-Up-Link
no switchport
ip address 100.100.10.6 255.255.255.252
no shutdown
exit
interface vlan 2
description VLAN-02-LAB-01
in address 192.168.2.3 255.255.255.0
```

```
standby 2 preempt
standby 2 priority 100
standby 2 ip 192.168.2.1
no shutdown
exit
interface vlan 3
description VLAN-03-LAB-02
ip address 192.168.3.3 255.255.255.0
standby 2 preempt
standby 2 priority 100
standby 2 ip 192.168.3.1
no shutdown
exit
interface vlan 4
description VLAN-04-LAB-03
ip address 192.168.4.3 255.255.255.0
standby 2 preempt
standby 2 priority 100
standby 2 ip 192.168.4.1
no shutdown
exit
ML-SW-02(config)#
ML-SW-02(config)#ip routing
ML-SW-02(config)#ip route 0.0.0.0 0.0.0.0 100.100.10.5
ML-SW-02(config)#exit
ML-SW-02#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ML-SW-02#
```

ISP

Router>enable

Router#conf t

Router(config)#hostname IPS

IPS(config)#interface gigabitEthernet 0/0/0

IPS(config-if)#description ML-SW-01-Link

IPS(config-if)#ip address 100.100.10.1 255.255.255.252

IPS(config-if)#no shutdown

IPS(config-if)#exit

IPS(config)#interface gigabitEthernet 0/0/1

IPS(config-if)#description ML-SW-02-Link

IPS(config-if)#ip address 100.100.10.5 255.255.255.252

IPS(config-if)#no shutdown

IPS(config-if)#exit

IPS(config)#interface gigabitEthernet 0/0/2

IPS(config-if)#description ISP-SW-Link

IPS(config-if)#ip address 8.8.8.1 255.255.255.0

IPS(config-if)#no shutdown

IPS(config-if)#exit

IPS(config)#ip route 0.0.0.0 0.0.0.0 100.100.10.2

IPS(config)#ip route 0.0.0.0 0.0.0.0 100.100.10.6

IPS(config)#exit

IPS#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

IPS#